

3.8.2 Externally Marked Route X (PSV/EMPA)

The practical and investigative skills will be assessed through:

- Practical Skills Verification (PSV) and
- Externally Marked Practical Assignment (EMPA).

The PSV requires teachers to verify their candidates' ability to demonstrate safe and skilful practical techniques and make valid and reliable observations.

The EMPA has two stages where candidates:

- Undertake a practical activity
- Complete a written EMPA test.

There are two windows of assessment for the EMPA:

- one for practical work (Section A: Task 1 and Task 2)
- one for the written test (Section B).

Each stage of the EMPA must be carried out

- under controlled conditions
- within the windows of assessment stipulated by AQA in the Instructions for Administration of the EMPA.

All students at a centre must complete the written test in a single uninterrupted session on the same day.

The EMPA is set and marked by AQA. Only one EMPA at each of AS and A2 will be provided in a given academic year.

Practical Skills Verification

Candidates following this route must undertake specific practical exercises. They will be required to work individually and carry out 5 short practical exercises under supervision in the laboratory during normal class time. The exercises will be set by AQA and may be undertaken at any stage during the course at the centre's discretion either as individual exercises or by organising more than one exercise to be taken at a said time. The candidates should be supervised during the practical work. They will not be expected to spend more than

3 hours in total of laboratory time in completing these exercises. The exercises will be typical of the normal practical work that would be expected to be covered as part of any AS or A2 physics course and should not add any additional burden to centres.

The teacher will confirm on the front cover of the written test, for each candidate that this requirement has been met. Failure to complete the tick box will lead to a mark of zero being awarded to the candidate for the whole of this unit. Knowledge and understanding of the skills shown in the tasks may be assessed of the EMPA written tests.

ICT

Candidates may use ICT where appropriate in the course of developing practical skills, for example in collecting and analysing data.

Externally Marked Practical Assignment (EMPA)

The Externally Marked Practical Assignment carries 55 marks and has two stages.

Stage 1: Collection and Processing of data

Candidates carry out practical work following AQA instructions. These will be laid out in Section A EMPA test answer booklet. The activity may be conducted in a normal timetabled lesson and at a time convenient to the centre but must be under controlled conditions and during the window of assessment for practical work. Candidates collect raw data and represent it in a table of their own design or make observations. The candidates' work must be handed to the teacher at the end of each session.

The activity will be made up of two tasks, centred around a particular area of physics. The tasks will assess the skills stipulated in the assessment objective AO3 (see section 4.2).

Centres will be guided how to set up the EMPA task by Teachers Notes which may be used, as described, or centres may make minor suitable modifications to materials or equipment following AQA guidelines. Any modifications made to the tasks must be indicated with the material sent to the examiner.

Candidates should work individually and be supervised throughout. The task will provide them with sufficient information to obtain reliable measurements which they will be required to identify, record, and process and eliminate possible anomalies and minimise measurement errors. They will be expected to then further analyse and evaluate their measurements in Stage 2. The questions in Section B of the EMPA will focus on both tasks.

There is no specified time limit for this stage.

Stage 2: The EMPA written test

The EMPA test should be taken under controlled conditions and during the window of assessment for the written test. All students at a centre must complete the written test in a single uninterrupted session on the same day. Each candidate is provided with a test paper (Section B of the EMPA) and the candidate's completed material written from Stage 1.

The test will be a duration of 1 hour 15 minutes.

Candidates will be required:

- to use their results and graph from Stage 1 to perform further analysis in order to arrive at a quantifiable outcome or conclusion
- to assess elements of the practical activity, such as the overall accuracy of the outcomes.

Use of ICT during the EMPA

ICT may be used during the EMPA Stages 1 and 2 but teachers should note any restrictions in the Teachers' Notes. Use of the internet is not permitted.

Candidates absent for the practical work

A candidate absent for the practical work (Stage 1) should be given an opportunity to carry out the practical work before they sit the EMPA written test. This may be with another group or at a different time. In extreme circumstances, when such arrangements are not possible the teacher can supply a candidate with class data. This must be noted on the Candidate Record Form, in this case the candidate cannot be awarded marks for Stage 1, but can still be awarded marks for Stage 2 of the assessment.

Material from AQA

For each EMPA AQA will provide:

- Teachers' Notes

- Section A and Section B papers of the EMPA test (Stage 1 and Stage 2 documentation).

When received, this material must be kept under secure conditions. Further details regarding this material will be provided.

Security of assignments

Completed EMPAs should be treated like examination papers and kept under secure conditions until sent to the AQA Examiner. All other EMPA materials should be kept under secure conditions until publication of results.

General Information

Route X

Administration

Only one EMPA will be available in any year at AS and at A2. AQA will stipulate a window of assessment during which the EMPA (task and test) must be completed.

Candidates may make only one attempt at a particular EMPA and redrafting is not permitted at any stage during the EMPA.

Work to be submitted

The material to be submitted to the examiner for each candidate consists of:

- the candidate's data in the Section A test papers (Stage 1 of the EMPA)
- the candidate's completed Section B test paper (Stage 2 of the EMPA) which includes the Candidate Record Form, including the PSV verification of the 5 practical exercises.

In addition each centre must provide:

- a Centre Declaration Sheet
- Details of any agreed amendments to the tasks, with information supporting the changes from the AQA Assessment Adviser.

Working in groups

For the PSV candidates may work in groups provided that any skills being assessed are the work of individual candidates. For the EMPA further guidance will be provided but the opportunity for group work will not be a common feature.

Other information

Section 6 of this specification outlines further guidance on the supervision and authentication of Internally assessed units.

Further support

AQA supports centres in a number of ways:

- A Teaching and learning resources web page which includes further information and guidance
- Assessment Advisers are appointed by AQA to provide advice on internally assessed units. Every centre is allocated an Assessment Adviser.

The Assessment Advisers can provide guidance on issues relating to the carrying out of tasks for assessment.

Any amendments to the EMPA task sheet must be discussed with the AQA Assessment Adviser and confirmation of the amendments made must be submitted to the AQA Examiner.